Finite-Prefix Terras Parity-Vector Stopping-Time Density Bounds in Lean 4

Benjamin Stanley Frohman
Independent Researcher | ORCID: 0009-0006-7068-3718
frohmanbenjamin@gmail.com

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Abstract

We present a zero-sorry, 7-module Lean 4 formalization (TerrasDensity/Basic.lean, Covering.lean, Density.lean, Inclusion.lean, Tail.lean) of the finite-prefix Terras parity-vector stopping-time density structure for the $3n+1$ map. Here FiniteStoppingTime n denotes the existence of $k \geq 1$ such that $\mathrm{iter}(k, n) < n$. The module Tail.lean formalizes the binomial tail count of bad parity words of length k, and Density.lean links residue class inclusion to the asymptotic density-one stopping-time statement, while strictly separating Terras density-one finite stopping time from the open global Collatz convergence claim ($\forall n > 0$, $\exists k$, $T^k(n) = 1$).

1 Architecture and Provenance

- **Origin Commit Timestamp:** Sunday, October 4, 2026, 15:45:31 CDT (2026-10-04T20:45:31Z).
- **Modules:** TerrasDensity.lean, TerrasDensity/Basic.lean, TerrasDensity/Covering.lean, TerrasDensity/Density.lean, TerrasDensity/Inclusion.lean, TerrasDensity/Tail.lean.
- **Honest Scope:** Proves the structural Terras parity-vector inclusion and binomial tail bounds with zero sorry. Does not claim that density-one finite stopping time implies that every positive integer reaches 1.

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